

Spine — Anatomy, Disc Prolapse & Infections

The Big Picture

Three blocks: **anatomy** of the vertebral column + disc, **disc prolapse** (cervical & lumbar), and **spinal infections** (pyogenic + TB/Pott's). Recurring exam hits: the 4-grade disc herniation scale, **cauda equina** = surgical emergency, and **S aureus** as top spinal-infection bug.

Anatomy Essentials

- **33 vertebrae** — 7 cervical, 12 thoracic, 5 lumbar, 5 sacral, 3–4 coccygeal.
- **Functions** — support/transmit weight to pelvis, protect cord, allow movement, muscle attachment.
- **Curves** — cervical lordosis (~40°), thoracic kyphosis (20–40°), lumbar lordosis (30–60°, ~2× thoracic). **Sagittal balance > coronal balance.**
- **Cervical quirks** — **8 cervical nerves** for 7 vertebrae (exit *above* their numbered vertebra; C8 between C7–T1). ~50% flexion-extension at occiput–C1; ~50% rotation at C1–C2 (odontoid/dens). **Vertebral artery** enters transverse foramen at **C6** up to C1.
- **Cauda equina** arises from conus medullaris (~T10–L1).

Intervertebral Disc

- Hydrostatic load-bearer: **nucleus pulposus** (type II collagen + hydrophilic proteoglycan, 70–90% water) inside **annulus fibrosus** (>60 lamellae of type I collagen, helicoid).
 - **Avascular** — nutrition by diffusion through end-plate. **L4-5 = largest avascular structure in body.**
 - **Posterior longitudinal ligament is weak** → most prolapses go **posterolateral**.
 - Facet joints = synovial, richly innervated, carry **18% of lumbar load**; degeneration = **spondylosis**.
 - Ageing: ↓ proteoglycan → water loss → ↓ disc height → uneven annular stress.
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Cervical Disc Prolapse

- **Who** — male, **30–50 yrs**, smokers; sudden flexion + twisting. Common at C5-6, C6-7.
- **Classification** — Site / Direction (posterolateral) / Amount: **1 Bulge** → **2 Protrusion** → **3 Extrusion** → **4 Sequestration**.
- **Clinical** — depends on what's compressed: root (radiculopathy), cord (**myelopathy**: loss of dexterity, gait instability, **Babinski/Hoffman** signs).
- **Imaging** — **MRI diagnostic**; CT ± myelography if MRI unsuitable.
- **Treatment** — conservative (collar/rest, analgesics, NSAID, **epidural steroid** max 3/yr, traction). **Surgery if acute myelopathy or progressive deficit** → ACDF (discectomy + fusion), disc arthroplasty (CDR), posterior microdiscectomy.

Lumbar Disc Prolapse

- **Where** — **L4-L5, L5-S1** (most stress). **3rd–4th decade**. Flexion + compression / heavy lift.
- Same 4-grade amount scale; **big posterior prolapse** → **cauda equina**.
- **Clinical** — sharp pain radiating back/buttock **below knee (sciatica)**; L4-5 prolapse hits **L5 root** (root affected = level below). Worse with **cough/Valsalva, sitting/driving** (↑ intradiscal pressure).

- **Exam** — sciatic list, **SLR/straight-leg raise**, bowstring, crossed sciatic, femoral stretch, Schober.
- **Imaging** — MRI (but **30% of asymptomatic people have abnormal MRI** — treat the patient, not the scan).
- **Treatment** — **90% improve conservatively** (rest, NSAID, PT, epidural/root block) over **6 weeks** before escalating.
- **Absolute surgical indications: cauda equina (operate ASAP), severe progressive motor deficit.**
- Relative: failed conservative care, intractable radicular pain, large extruded fragment, recurrence.
- Options: open/microdiscectomy, endoscopic, ± fusion. Microdiscectomy = faster recovery but same outcome by 8–12 wks.
- Results: 85–95% good short-term → 55–70% long-term; back pain = main dissatisfaction.

Spinal Infections (Pyogenic)

- **Spondylodiscitis levels** — lumbar **59%** > thoracic 33% > cervical 8%; epidural abscess in ~7%.
- **Risk** — peak **7th decade**; diabetes, immunosuppression, steroids, malnutrition, UTI, IVDU, invasive procedures.
- **Organism** — **S aureus 30–50%** (commonest); gram-negatives from UTI/chest/skin.
- **Spread** — haematogenous, direct, post-op.
- **Clinical** — pain + focal tenderness **90%**, fever 61%, root signs 60%, muscle spasm, meningism.
- **Investigations** — WBC raised in <½; **ESR >50 in 80–90%** + CRP (track treatment); blood culture 20–25%; ASO/brucella/salmonella serology; **biopsy** (open >80% accurate). **MRI = most important (>95%)**: T2 ↑, T1 ↓, ring enhancement, end-plate destruction.
- **Treatment** — **antibiotics by sensitivity, IV then oral, >6 weeks** (continue 1 month after ESR/symptoms normalise) + bracing/rest. **Surgery if**: conservative fails, organism unconfirmed, neuro deficit/epidural spread, instability, **abscess/sequestra** → drain, decompress, bone graft + fusion.

Spinal TB (Pott's Disease)

- Commonest **granulomatous** spinal infection (with brucellosis, fungi).
- **Pott's paraplegia = most feared complication.** **Early-onset** paresis = pressure (abscess/caseous/sequestrum); **late-onset** = deformity, reactivation, or cord vascular insufficiency.
- **X-ray** — x-ray whole spine (skip lesions); early = local osteoporosis of 2 adjacent vertebrae + disc-space narrowing → later **angular (kyphotic) deformity**; paravertebral abscess (may calcify on healing).
- **Investigations** — ESR, **Mantoux/PCR**, needle biopsy. DDx: pyogenic, malignancy.
- **Treatment** — **anti-TB chemotherapy as effective as surgery** for arresting disease. Operate for drainable abscess, marked bone destruction, severe kyphosis/paraparesis, instability, sequestra → **anterior approach, excise + rib strut graft** ± posterior fixation; chemo continues.

Walk-Away Points

- Disc herniation amount: **Bulge** → **Protrusion** → **Extrusion** → **Sequestration**; PLL weak → posterolateral.

- Lumbar prolapse hits the root **below** (L4-5 → L5); **cauda equina = operate ASAP**.
- **90% of lumbar prolapse settle conservatively** in 6 wks; treat patient not MRI (30% false positives).
- Spinal infection: **lumbar commonest, S aureus top bug, MRI best test, antibiotics >6 weeks**.
- **Pott's** = TB spine; paraplegia is feared; anti-TB chemo first, rib strut graft for surgery.

Spine (Anatomy / Disc / Infection) — Revision Table	
Anatomy	Detail
Vertebrae	7C, 12T, 5L, 5S, 3–4 coccygeal
Curves	Cervical lordosis 40°, thoracic kyphosis 20–40°, lumbar lordosis 30–60°
Cervical nerves	8 nerves , exit above numbered vertebra; vertebral artery enters C6
Disc	NP (type II, 70–90% water) + annulus (type I, >60 lamellae); avascular
Key fact	PLL weak → posterolateral prolapse ; facets carry 18% lumbar load
Disc herniation grade	Meaning
1 Bulge	Diffuse
2 Protrusion	Focal, annulus intact
3 Extrusion	Through annulus
4 Sequestration	Free fragment
Cervical disc prolapse	Detail
Who	Male, 30–50, smoker; C5-6, C6-7
Signs	Myelopathy: Babinski, Hoffman , gait/dexterity loss
Imaging	MRI diagnostic
Surgery	ACDF, CDR; if acute myelopathy/progressive deficit
Lumbar disc prolapse	Detail
Where	L4-5, L5-S1 ; 3rd–4th decade
Root	Affects root below (L4-5 → L5)
Worse with	Cough/Valsalva, sitting, driving
Tests	SLR, bowstring, femoral stretch
Conservative	90% improve , 6 wks; treat patient not MRI (30% false +)
Absolute surgery	Cauda equina (ASAP) , progressive motor deficit
Pyogenic spinal infection	Detail
Levels	Lumbar 59% > thoracic 33% > cervical 8%
Organism	S aureus 30–50%
Risk	7th decade, diabetes, immunosuppression, IVDU
Clinical	Pain/tenderness 90%, fever 61%
Investigations	ESR>50 (80–90%) , CRP, biopsy; MRI best (>95%)
Treatment	Abx by sensitivity IV→oral >6 wks ; surgery if abscess/sequestra/deficit/instability
Pott's (spinal TB)	Detail
Type	Commonest granulomatous
Feared	Pott's paraplegia (early = pressure; late = deformity/vascular)
X-ray	2 adjacent vertebrae osteoporosis + disc narrowing → angular kyphosis
Dx	Mantoux/PCR, biopsy
Treatment	Anti-TB chemo (= surgery efficacy); anterior excision + rib strut graft

Bold = high-yield.